t3-w04 — condition C1

Three Dimensional Shapes — Three dimensional figures: properties of cubes, cuboids, pyramids and cones

> The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

entry_behaviour

• Students have been taught the differences between cube and cuboid.

instructional_materials

• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.

Objective 1 — Identify the properties of cube and cuboids

objective_text

Identify the properties of cube and cuboids

current_step

1

instructional_events

• present_content

assessment_points

• How many vertices, has a cuboid?
• How many edges has a cube?
• Draw a cube and cuboid then label.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — Identify the properties of pyramids and cones

objective_text

Identify the properties of pyramids and cones

current_step

2

instructional_events

• present_content

assessment_points

• Draw an hexagonal base pyramid and list the properties.
• How many vertices has a tetrahedron.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Properties of Cubes and Cuboids

> EXPLANATION
> A **cube** is a special kind of cuboid where all six faces are squares. This means:
> - 6 faces, each a square
> - 12 edges, all the same length
> - 8 vertices (corners)
>
> A **cuboid** (also called a rectangular prism) has faces that are rectangles. Its properties are:
> - 6 faces (pairs of opposite faces are equal rectangles)
> - 12 edges, but opposite edges are equal in length (there are three different edge lengths: length, width, height)
> - 8 vertices
>
> Because a cube is just a cuboid with all edges equal, you can think of a cube as a “regular” cuboid.
>
> When you look at a shape, count the faces, edges, and vertices to identify it.

> WORKED_EXAMPLE
> **Worked Example:**
>
> Identify the shape shown below (imagine a drawing of a rectangular box where the length is longer than the width and height).
>
> - Faces: 6 (three pairs of opposite rectangles)
> - Edges: 12 (4 edges of one length, 4 of another, 4 of the third)
> - Vertices: 8
>
> Since the faces are rectangles and not all edges are the same length, this is a **cuboid**, not a cube.

1. How many vertices does a cuboid have?
answer: 8

2. How many edges does a cube have?
answer: 12

3. Draw a cube and a cuboid, then label the number of faces, edges, and vertices for each.
answer: Cube: 6 faces, 12 edges, 8 vertices; Cuboid: 6 faces, 12 edges, 8 vertices (but edges have three different lengths).

Objective 2

Properties of Pyramids and Cones

> EXPLANATION
> A **pyramid** is a solid that has a polygonal base and triangular faces that meet at a single point called the apex. The number of triangular faces equals the number of sides of the base. Key properties:
> - **Base**: any polygon (triangle, square, pentagon, etc.).
> - **Vertices**: one vertex at each corner of the base plus the apex. So a pyramid with an n-sided base has n + 1 vertices.
> - **Edges**: the edges of the base plus one edge from each base vertex to the apex, giving n + n = 2n edges.
> - **Faces**: the base plus n triangular faces, total n + 1 faces.
>
> A **cone** is a solid with a circular base and a single curved surface that narrows to a point called the apex. Its properties are:
> - **Base**: one circle.
> - **Vertices**: only the apex (the circle has no vertices).
> - **Edges**: none (the circular edge of the base is not counted as a straight edge).
> - **Faces**: one curved surface plus the circular base, so 2 faces.
>
> Remember: the shape of the base determines how many triangular faces a pyramid has, while a cone always has a circular base.

> WORKED_EXAMPLE
> **Worked example:** Identify the properties of a square-based pyramid.
> - Base: square (4 sides).
> - Vertices: 4 base vertices + 1 apex = 5 vertices.
> - Edges: 4 base edges + 4 side edges = 8 edges.
> - Faces: 1 square base + 4 triangular faces = 5 faces.
> So a square-based pyramid has 5 vertices, 8 edges, and 5 faces.

1. Draw a hexagonal base pyramid and list its number of vertices, edges, and faces.
answer: Vertices: 7 (6 base + 1 apex); Edges: 12 (6 base + 6 side); Faces: 7 (1 hexagon base + 6 triangles).

2. How many vertices does a tetrahedron have?
answer: 4 vertices.

3. A cone has a circular base. State how many vertices, edges, and faces it has.
answer: Vertices: 1 (the apex); Edges: 0; Faces: 2 (one circular base and one curved surface).

4. A pyramid has a triangular base. How many faces does it have?
answer: 4 faces (1 triangular base + 3 triangular sides).

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Method written from event labels — the shape of the teaching without its substance	**substituted**
Worked example a problem with its solution worked through	2	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	7	Assessment written from the teacher's evaluation questions instead of her set work	**substituted**
	7	Assessment	**substituted**

Answer what the learner checks against	checked against the evaluation answers rather than the taught method